UNIT 2 · TOPIC 11(B)

Operation of Farm Equipment — Adjustments, Calibration and Field Patterns

Syllabus topic: Construction and operation of farm equipment

Setting the machine — mould-board plough adjustments

Adjustment	Value / purpose (eCourse)
Vertical (down) suction	3–5 mm — vertical clearance at the share–landside joint; makes the plough penetrate to depth
Horizontal (side) suction	5–13 mm — lateral clearance at the landside heel; holds the width of the furrow slice
Angle of cutting edge	40–45° to the direction of travel
Throat clearance	Perpendicular distance from share point to the underside of the beam — room for trash and soil flow
Vertical clevis	Holes at the beam end — set depth and the line of pull
Horizontal clevis	Lateral adjustment of the plough relative to the line of draft
Cross-shaft (tractor plough)	Rotated to increase or decrease the width of the first furrow
Coulter	Vertical: ½ of ploughing depth or 50 mm; horizontal: 20 mm outside the landside; diameter 30–45 cm
Jointer	Depth 4–5 cm
Rear furrow wheel	6–12.5 mm clearance between the rear of the landside and the furrow wall

- **Plough size** = perpendicular distance from the share wing to the line joining share point and landside heel: tractor ploughs commonly 30, 35 or 40 cm; animal-drawn 13–20 cm.
- Compare: Unit 6, Topic 4 (Kepner) gives suction as ⅛–⅜ in (≈ 3–10 mm) — the same order.

Scouring and overload protection in operation

- A tool scours (soil slides off without sticking) when soil–metal friction is lower than soil–soil friction. Non-scouring soil rides on a soil layer stuck to the tool, raising draft.
- Remedies: smooth, non-wetting surfaces — Teflon-coated mould boards gave **23%** less draft at 5.6 km/h in tests; slat mould boards; enamel coatings; air-lubricated or belt-type surfaces.
- Mounted and semi-mounted ploughs need individual overload protection at higher speeds: **spring-trip** (bottom swings back over an obstruction and resets) or **hydraulic-reset** devices.

Calibrate before work

1. Find the nominal width W = number of furrow openers × spacing.
2. For 1 ha, strip length L = 10,000/W; revolutions of the ground wheel N = 10,000/(πD × W).
3. Jack the drill, mark the drive wheel, fill the hopper and place a container under each boot.
4. Turn the wheel N times (or a convenient fraction), weigh the seed and calculate kg/ha.

5. Adjust the rate control and repeat until the desired rate is reached.

For planters, seed-to-seed distance = πD/(N·C) (N = plate revolutions per wheel revolution, C = cells on the plate); skid of the drive wheel widens it (Topic 6a). Full procedures: Unit 3, Topic 1.

> **Worked example (eCourse).** 10 furrows at 20 cm (W = 2 m), drive wheel 1.5 m, 500 revolutions, 20 kg collected → area = π × 1.5 × 2 × 500 = 4712.3 m² → seed rate = 20 × 10,000/4712.3 = **42.22 kg/ha**.

Field operation — patterns and efficiency

Objectives of a field pattern:

1. Minimum total field travel.
2. Minimum non-working turns.
3. Minimum non-working travel in the interior of the field.
4. A level surface that does not pond water.
5. Avoid repeated passes over the same ground (soil compaction — Unit 6, Topic 9).

- Tillage and seeding machines can make square turns; raking, windrowing and baling usually follow a rounded-corner pattern.
- Irregular fields give lower field efficiency than rectangular ones (more turning time).
- Field efficiency falls as theoretical field capacity rises (wider/faster machines lose proportionally more time in turns and servicing).

Where field time goes (count these when estimating capacity or cost):

- Machine preparation at the farmstead and travel to and from the field.
- Preparation in the field before and after work (daily servicing, preparing for towing).
- Theoretical field time (full width, optimum speed).
- Turning and crossing waterways; loading/unloading if not done on the go.
- Adjustments and unplugging; refuelling, lubrication, chain tightening; field repairs.

$$\text{TFC (ha/h)} = W \cdot S/10;\ \text{EFC} = W \cdot S \cdot E_f/10;\ \text{material capacity } M = S \cdot W \cdot E_f \cdot Y/10$$

> **Worked example (eCourse).** A machine 20 m wide at 5 km/h: TFC = 5 × 20/10 = **10 ha/h**. With 0.75 h lost in a 10-h day, E_f = (10 - 0.75)/10 = **92.5%** and EFC = 5 × 20 × 0.925/10 = **9.25 ha/h**. (More on field capacity: Unit 3, Topic 6a.)

Safe operation

- Hitch only at the drawbar or linkage points provided; keep guards in place on PTO shafts and drives (Unit 5).
- Lower or block raised implements before working under them; stop the PTO before adjusting or unclogging.
- Follow the operator's manual for settings, lubrication and service intervals (Unit 4, Topic 12).

Self-check

Question	Answer
What does vertical suction do, and how much is typical?	Makes the plough penetrate; ~3–5 mm
Which adjustment sets the width of the first furrow on a tractor plough?	The cross-shaft
Where is the coulter set horizontally?	About 20 mm outside the landside
Name two objectives of a field pattern.	Minimum non-working turns; minimum total travel (also: level surface, less compaction)
A 2-m drill sows 4712.3 m² and collects 20 kg. Seed rate?	42.22 kg/ha

Sources: ICAR eCourse "Farm Machinery and Equipment I" — course content by Dr Ritu Dogra (Dept. of Farm Machinery & Power Engg., PAU Ludhiana), reviewed by Dr K. K. Singh (ICAR) — Lessons 2 (materials of construction), 3 (heat treatment), 5–6 (field capacity, field patterns), 8–9 (mould-board plough construction, adjustments, operation) and 24 (calibration); published as an AgriMoon e-book.